

```
In [2]: import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [3]: df = pd.read_excel("Online_Retail.xlsx")
```

```
In [4]: df
```

Out[4]:

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID
0	536365	85123A	WHITE HANGING HEART T- LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850.0
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850.0
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850.0
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850.0
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	17850.0
...	...	...	...	...	...	...	...
541904	581587	22613	PACK OF 20 SPACEBOY NAPKINS	12	2011-12-09 12:50:00	0.85	12680.0
541905	581587	22899	CHILDREN'S APRON DOLLY GIRL	6	2011-12-09 12:50:00	2.10	12680.0
541906	581587	23254	CHILDRENS CUTLERY DOLLY GIRL	4	2011-12-09 12:50:00	4.15	12680.0
541907	581587	23255	CHILDRENS CUTLERY CIRCUS PARADE	4	2011-12-09 12:50:00	4.15	12680.0
541908	581587	22138	BAKING SET 9 PIECE RETROSPOT	3	2011-12-09 12:50:00	4.95	12680.0

541909 rows × 8 columns



```
In [5]: df.head()
```

Out[5]:

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Cou
0	536365	85123A	WHITE HANGING HEART T- LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850.0	Un King
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850.0	Un King
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850.0	Un King
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850.0	Un King
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	17850.0	Un King



```
In [6]: df.tail()
```

Out[6]:

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID
541904	581587	22613	PACK OF 20 SPACEBOY NAPKINS	12	2011-12-09 12:50:00	0.85	12680.0
541905	581587	22899	CHILDREN'S APRON DOLLY GIRL	6	2011-12-09 12:50:00	2.10	12680.0
541906	581587	23254	CHILDRENS CUTLERY DOLLY GIRL	4	2011-12-09 12:50:00	4.15	12680.0
541907	581587	23255	CHILDRENS CUTLERY CIRCUS PARADE	4	2011-12-09 12:50:00	4.15	12680.0
541908	581587	22138	BAKING SET 9 PIECE RETROSPOT	3	2011-12-09 12:50:00	4.95	12680.0



In [7]:

```
df.sample(7)
```

Out[7]:

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID
48546	540469	22023	EMPIRE BIRTHDAY CARD	12	2011-01-07 14:04:00	0.42	12484.0
425153	573305	23554	LANDMARK FRAME OXFORD STREET	2	2011-10-28 16:13:00	12.50	17837.0
476522	577023	21481	FAWN BLUE HOT WATER BOTTLE	8	2011-11-17 12:43:00	3.75	16186.0
516324	579885	23405	HOME SWEET HOME 2 DRAWER CABINET	2	2011-11-30 17:37:00	4.95	15444.0
211367	555368	22847	BREAD BIN DINER STYLE IVORY	2	2011-06-02 14:30:00	16.95	13437.0
441437	574573	22940	FELTCRAFT CHRISTMAS FAIRY	1	2011-11-04 16:35:00	4.25	14592.0
376871	569538	85152	HAND OVER THE CHOCOLATE SIGN	12	2011-10-04 15:55:00	2.10	14462.0



In [10]: `df.shape`

Out[10]: (541909, 8)

In [12]: `df.dtypes`

Out[12]: InvoiceNo object
StockCode object
Description object
Quantity int64
InvoiceDate datetime64[ns]
UnitPrice float64
CustomerID float64
Country object
dtype: object

In [13]: `df.columns`

```
Out[13]: Index(['InvoiceNo', 'StockCode', 'Description', 'Quantity', 'InvoiceDate',
               'UnitPrice', 'CustomerID', 'Country'],
              dtype='object')
```

```
In [14]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 541909 entries, 0 to 541908
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype
---  -
0   InvoiceNo        541909 non-null object
1   StockCode        541909 non-null object
2   Description      540455 non-null object
3   Quantity         541909 non-null int64
4   InvoiceDate      541909 non-null datetime64[ns]
5   UnitPrice        541909 non-null float64
6   CustomerID       406829 non-null float64
7   Country          541909 non-null object
dtypes: datetime64[ns](1), float64(2), int64(1), object(4)
memory usage: 33.1+ MB
```

```
In [15]: df.describe()
```

```
Out[15]:
```

	Quantity	InvoiceDate	UnitPrice	CustomerID
count	541909.000000	541909	541909.000000	406829.000000
mean	9.552250	2011-07-04 13:34:57.156386048	4.611114	15287.690570
min	-80995.000000	2010-12-01 08:26:00	-11062.060000	12346.000000
25%	1.000000	2011-03-28 11:34:00	1.250000	13953.000000
50%	3.000000	2011-07-19 17:17:00	2.080000	15152.000000
75%	10.000000	2011-10-19 11:27:00	4.130000	16791.000000
max	80995.000000	2011-12-09 12:50:00	38970.000000	18287.000000
std	218.081158	NaN	96.759853	1713.600303

```
In [16]: df.isnull().sum()
```

```
Out[16]: InvoiceNo      0
StockCode      0
Description    1454
Quantity       0
InvoiceDate    0
UnitPrice      0
CustomerID    135080
Country        0
dtype: int64
```

```
In [17]: df.duplicated().sum()
```

Out[17]: np.int64(5268)

```
In [18]: df=df.dropna(subset=['CustomerID'])
```

```
In [19]: df.shape
```

Out[19]: (406829, 8)

```
In [21]: df=df.drop_duplicates() # duplicate value removed
```

```
In [23]: df.shape
```

Out[23]: (401604, 8)

```
In [28]: df=df[df['Quantity']>0] #-ve qty removed
```

```
In [29]: df=df[df['UnitPrice']>0] #-ve price removed
```

```
In [32]: df['TotalAmount'] = df['Quantity'] * df['UnitPrice']
```

```
In [33]: df[['Quantity', 'UnitPrice', 'TotalAmount']].head()
```

Out[33]:

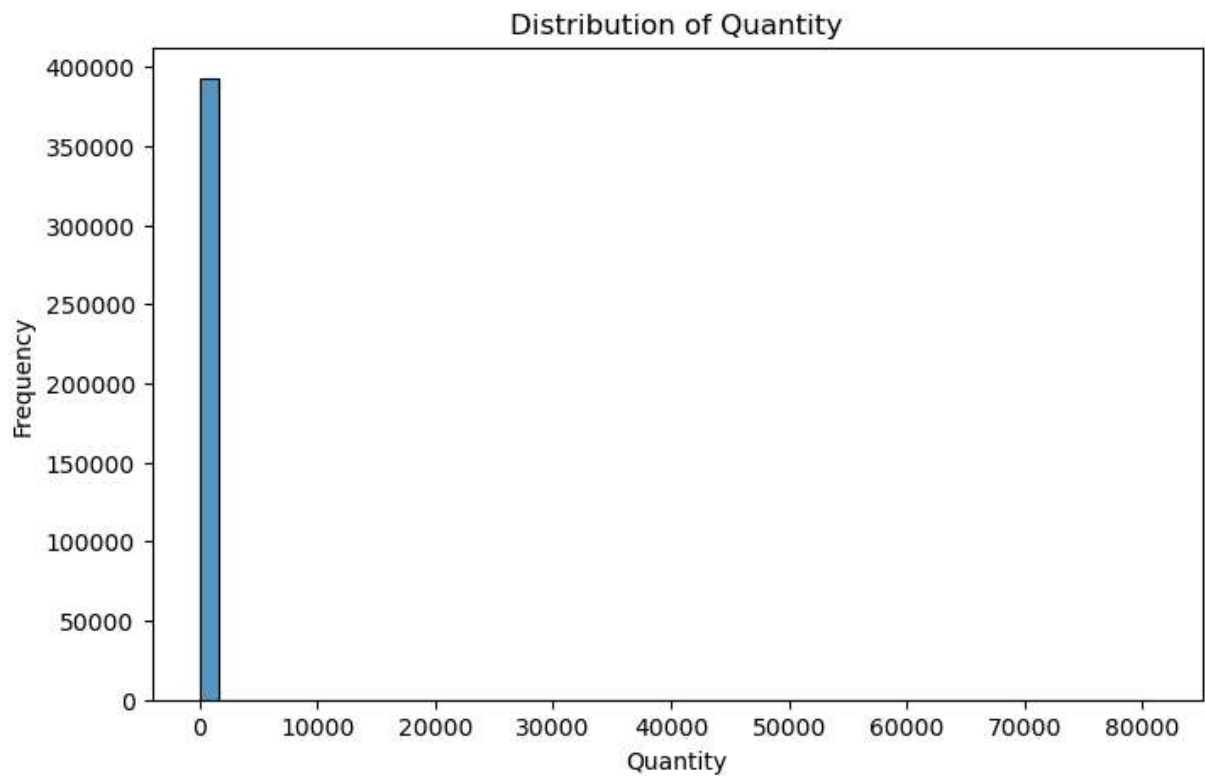
	Quantity	UnitPrice	TotalAmount
0	6	2.55	15.30
1	6	3.39	20.34
2	8	2.75	22.00
3	6	3.39	20.34
4	6	3.39	20.34

```
In [34]: plt.figure(figsize=(8,5))

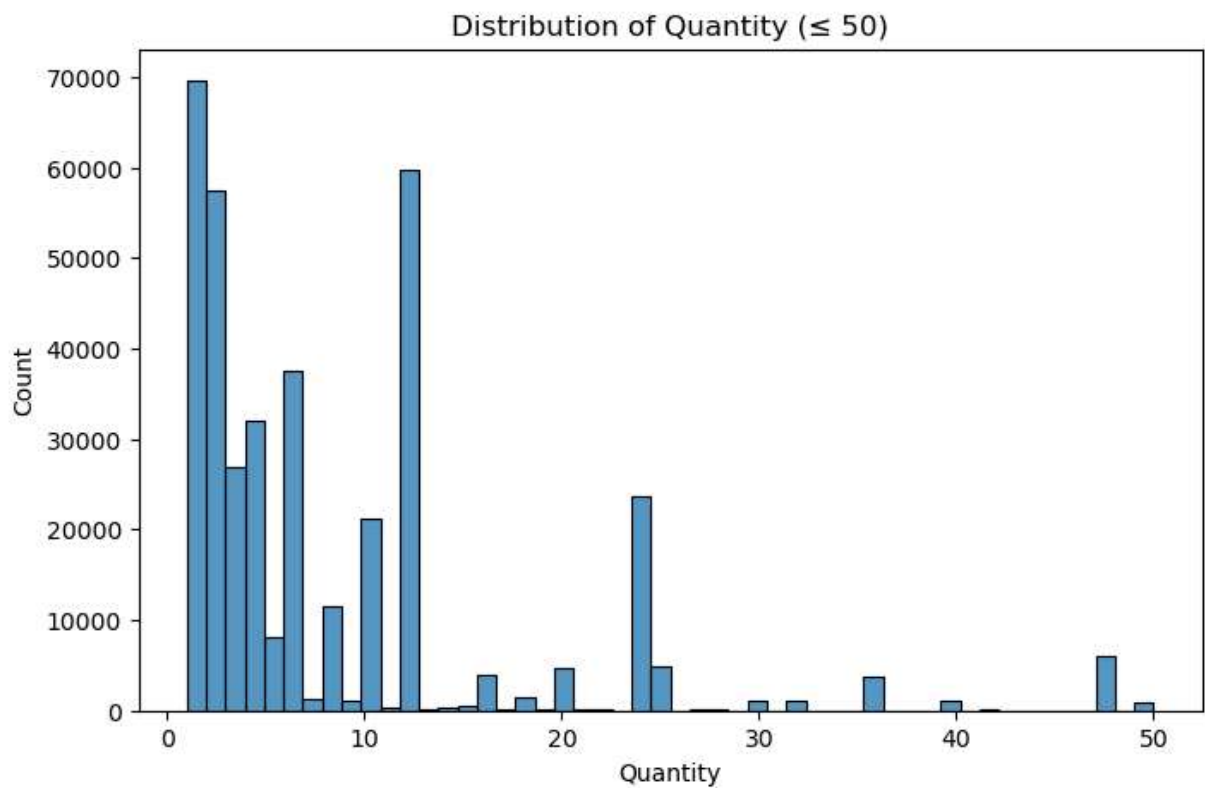
sns.histplot(df['Quantity'], bins=50)

plt.title("Distribution of Quantity")
plt.xlabel("Quantity")
plt.ylabel("Frequency")

plt.show()
```

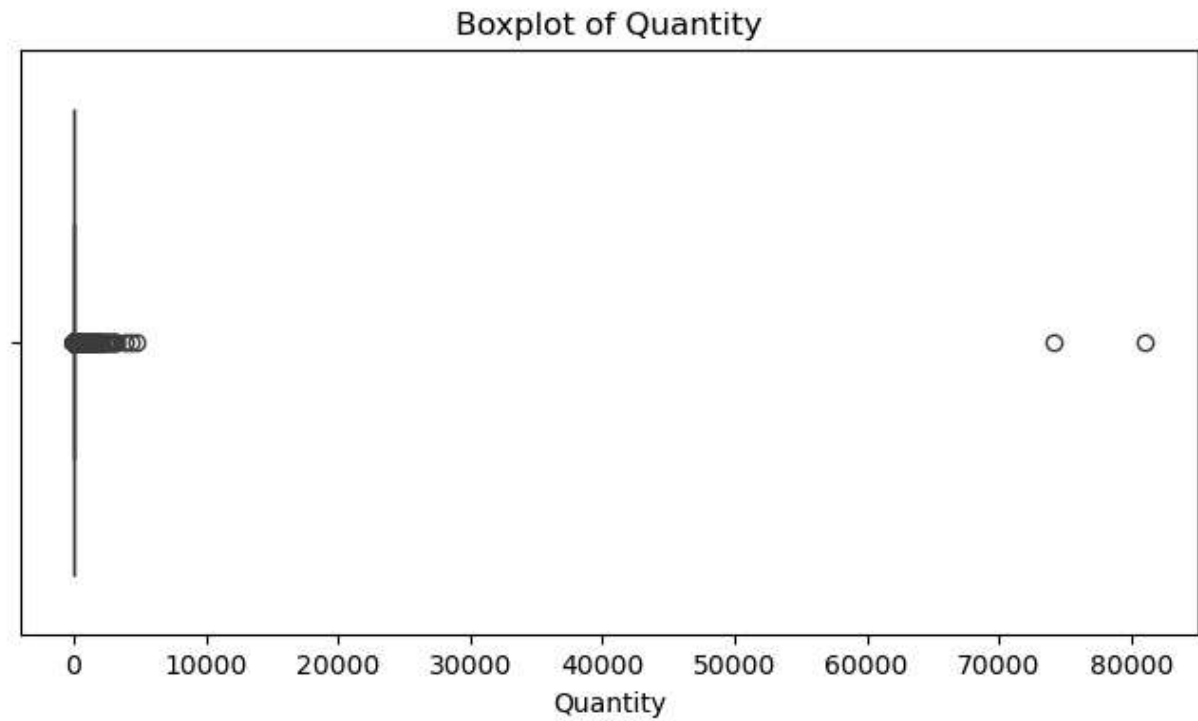


```
In [43]: plt.figure(figsize=(8,5))
sns.histplot(df[df['Quantity'] <= 50]['Quantity'], bins=50)
plt.title("Distribution of Quantity ( $\leq 50$ )")
plt.show()
```

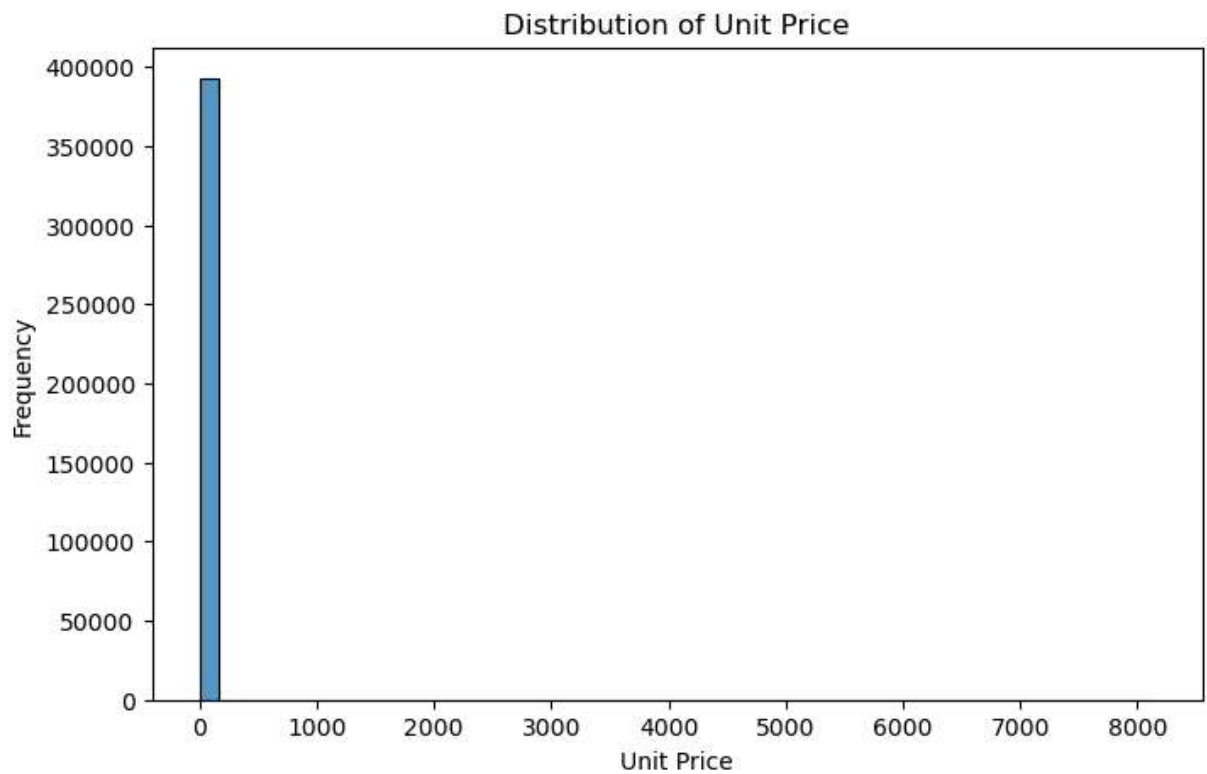


```
In [35]: plt.figure(figsize=(8,4))
```

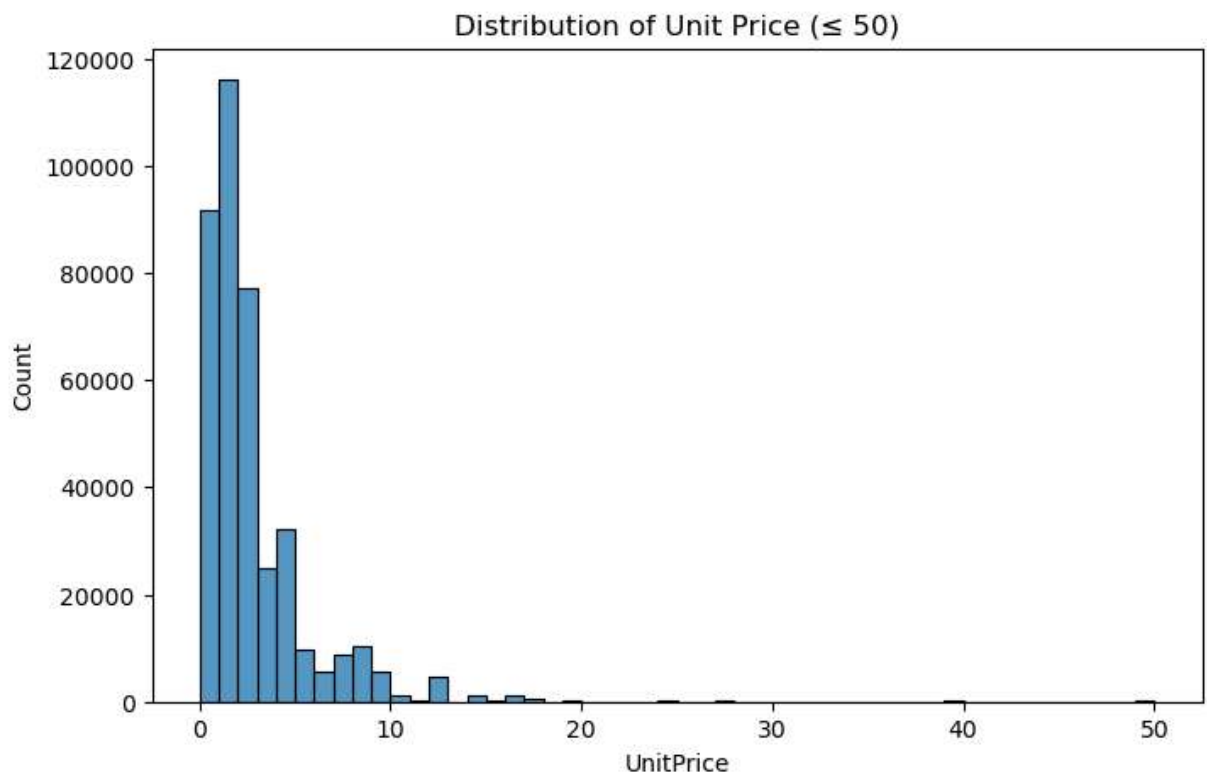
```
sns.boxplot(x=df['Quantity'])  
  
plt.title("Boxplot of Quantity")  
  
plt.show()
```



```
In [36]: plt.figure(figsize=(8,5))  
  
sns.histplot(df['UnitPrice'], bins=50)  
  
plt.title("Distribution of Unit Price")  
plt.xlabel("Unit Price")  
plt.ylabel("Frequency")  
  
plt.show()
```

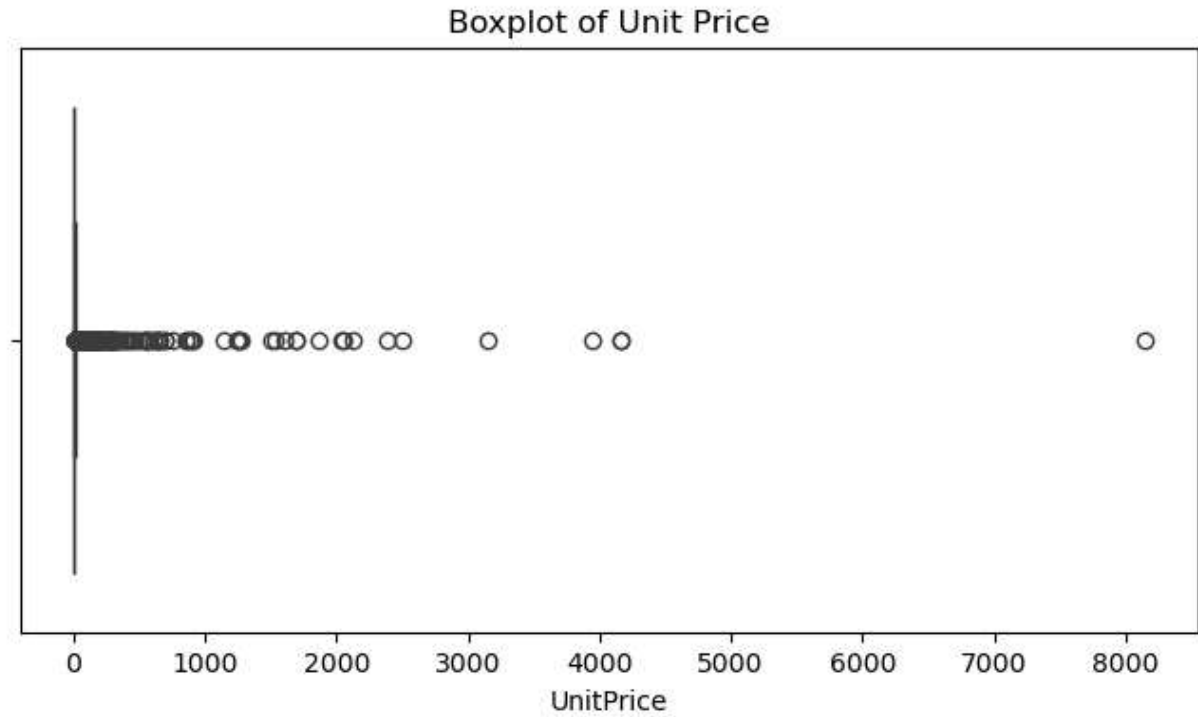


```
In [42]: plt.figure(figsize=(8,5))
sns.histplot(df[df['UnitPrice'] <= 50]['UnitPrice'], bins=50)
plt.title("Distribution of Unit Price ( $\leq 50$ )")
plt.show()
```

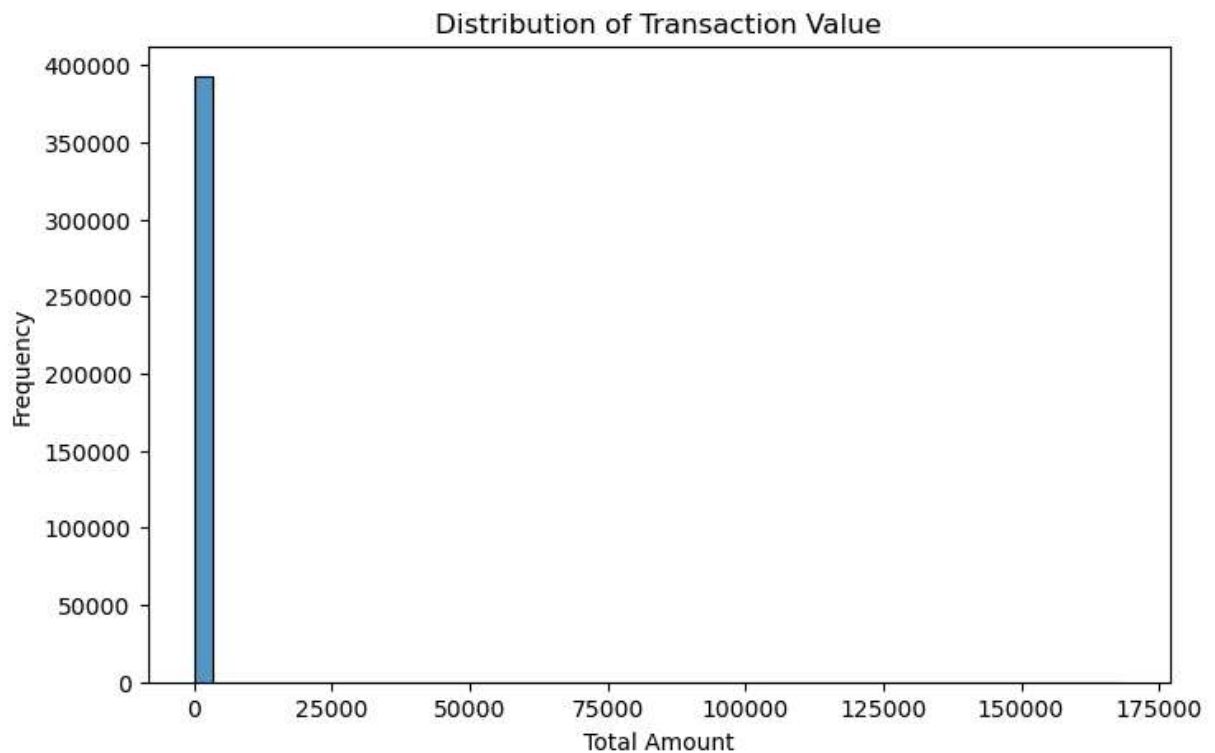


```
In [37]: plt.figure(figsize=(8,4))
```

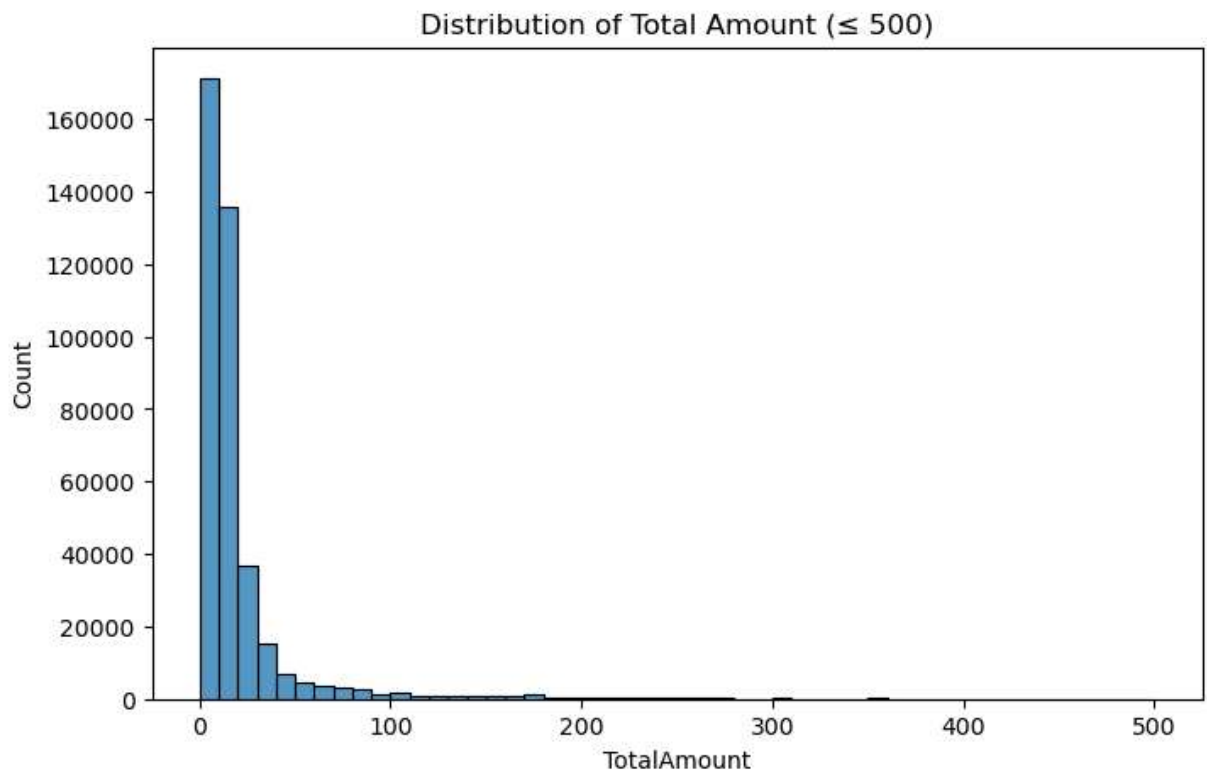
```
sns.boxplot(x=df['UnitPrice'])  
  
plt.title("Boxplot of Unit Price")  
  
plt.show()
```



```
In [38]: plt.figure(figsize=(8,5))  
  
sns.histplot(df['TotalAmount'], bins=50)  
  
plt.title("Distribution of Transaction Value")  
plt.xlabel("Total Amount")  
plt.ylabel("Frequency")  
  
plt.show()
```



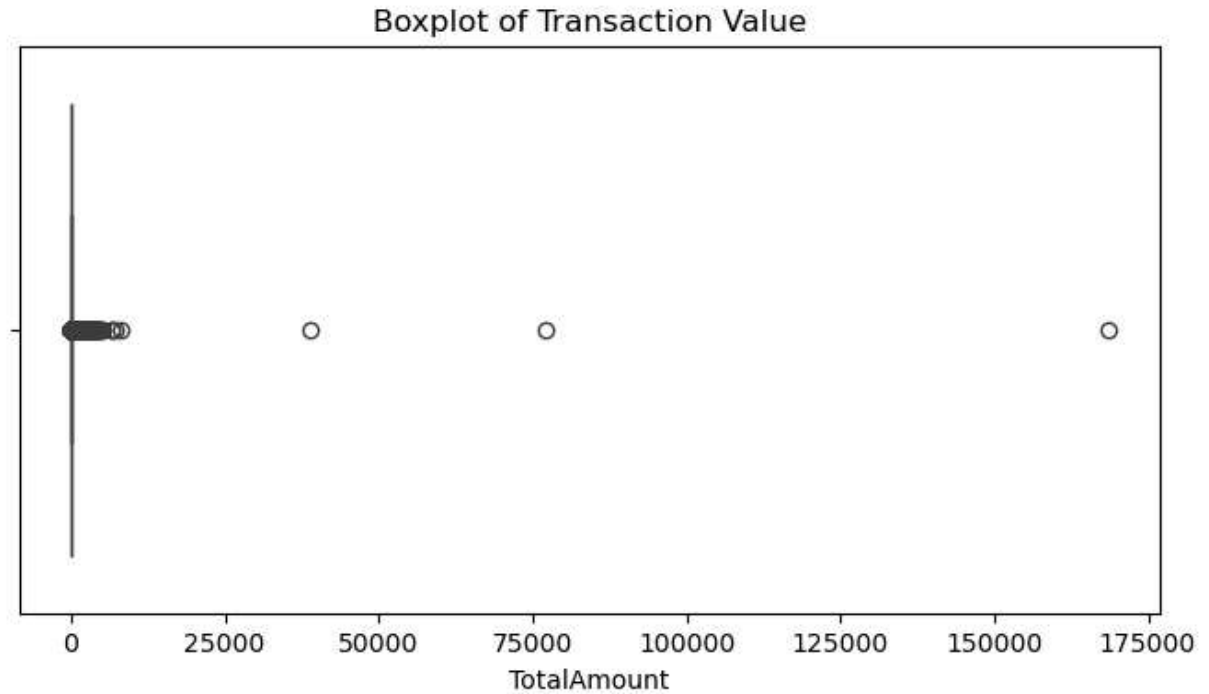
```
In [44]: plt.figure(figsize=(8,5))
sns.histplot(df[df['TotalAmount'] <= 500]['TotalAmount'], bins=50)
plt.title("Distribution of Total Amount ( $\leq 500$ )")
plt.show()
```



```
In [39]: plt.figure(figsize=(8,4))

sns.boxplot(x=df['TotalAmount'])
```

```
plt.title("Boxplot of Transaction Value")
plt.show()
```



```
In [45]: country_revenue = (
    df.groupby('Country')['TotalAmount']
        .sum()
        .sort_values(ascending=False)
    )

country_revenue.head(10)
```

```
Out[45]: Country
United Kingdom    7285024.644
Netherlands       285446.340
EIRE              265262.460
Germany          228678.400
France           208934.310
Australia        138453.810
Spain            61558.560
Switzerland      56443.950
Belgium          41196.340
Sweden           38367.830
Name: TotalAmount, dtype: float64
```

```
In [48]: plt.figure(figsize=(12,6))

country_revenue.head(10).plot(kind='bar')

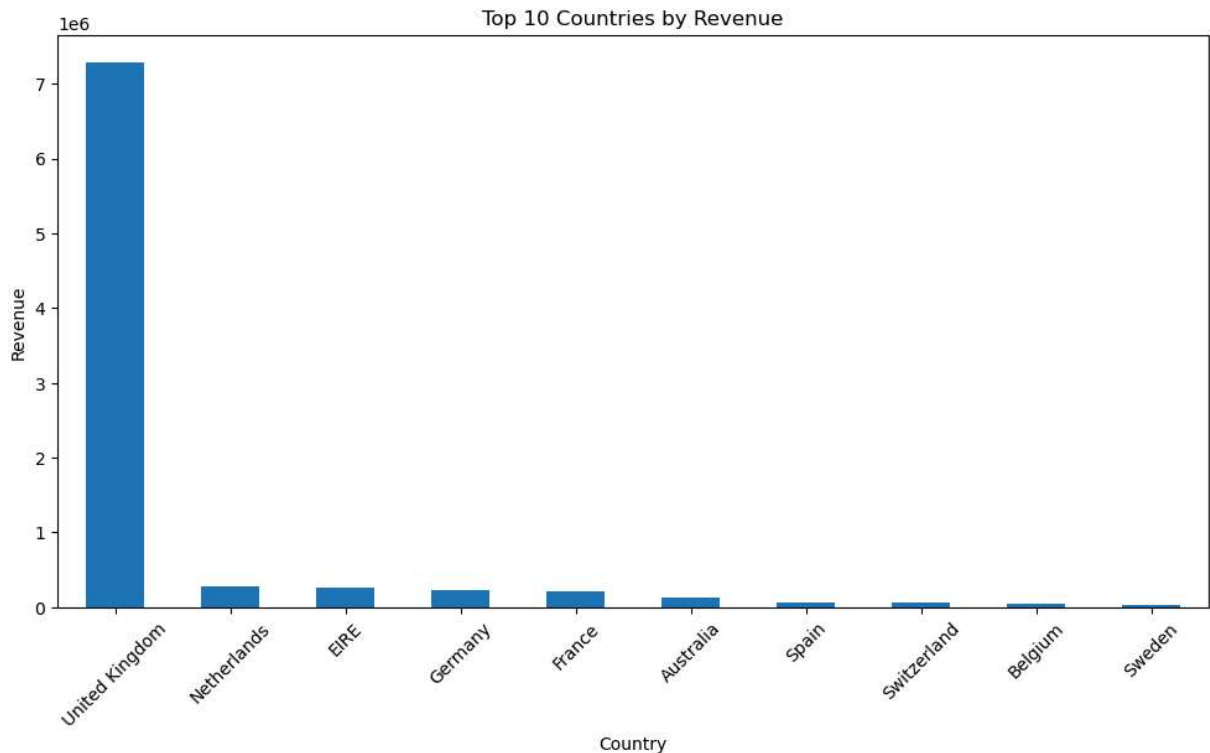
plt.title("Top 10 Countries by Revenue")

plt.xlabel("Country")
```

```
plt.ylabel("Revenue")

plt.xticks(rotation=45)

plt.show()
```



```
In [50]: product_revenue = (
    df.groupby('Description')['TotalAmount']
      .sum()
      .sort_values(ascending=False)
    )
product_revenue.head(10)
```

```
Out[50]: Description
PAPER CRAFT , LITTLE BIRDIE      168469.60
REGENCY CAKESTAND 3 TIER         142264.75
WHITE HANGING HEART T-LIGHT HOLDER 100392.10
JUMBO BAG RED RETROSPOT          85040.54
MEDIUM CERAMIC TOP STORAGE JAR   81416.73
POSTAGE                          77803.96
PARTY BUNTING                   68785.23
ASSORTED COLOUR BIRD ORNAMENT    56413.03
Manual                          53419.93
RABBIT NIGHT LIGHT               51251.24
Name: TotalAmount, dtype: float64
```

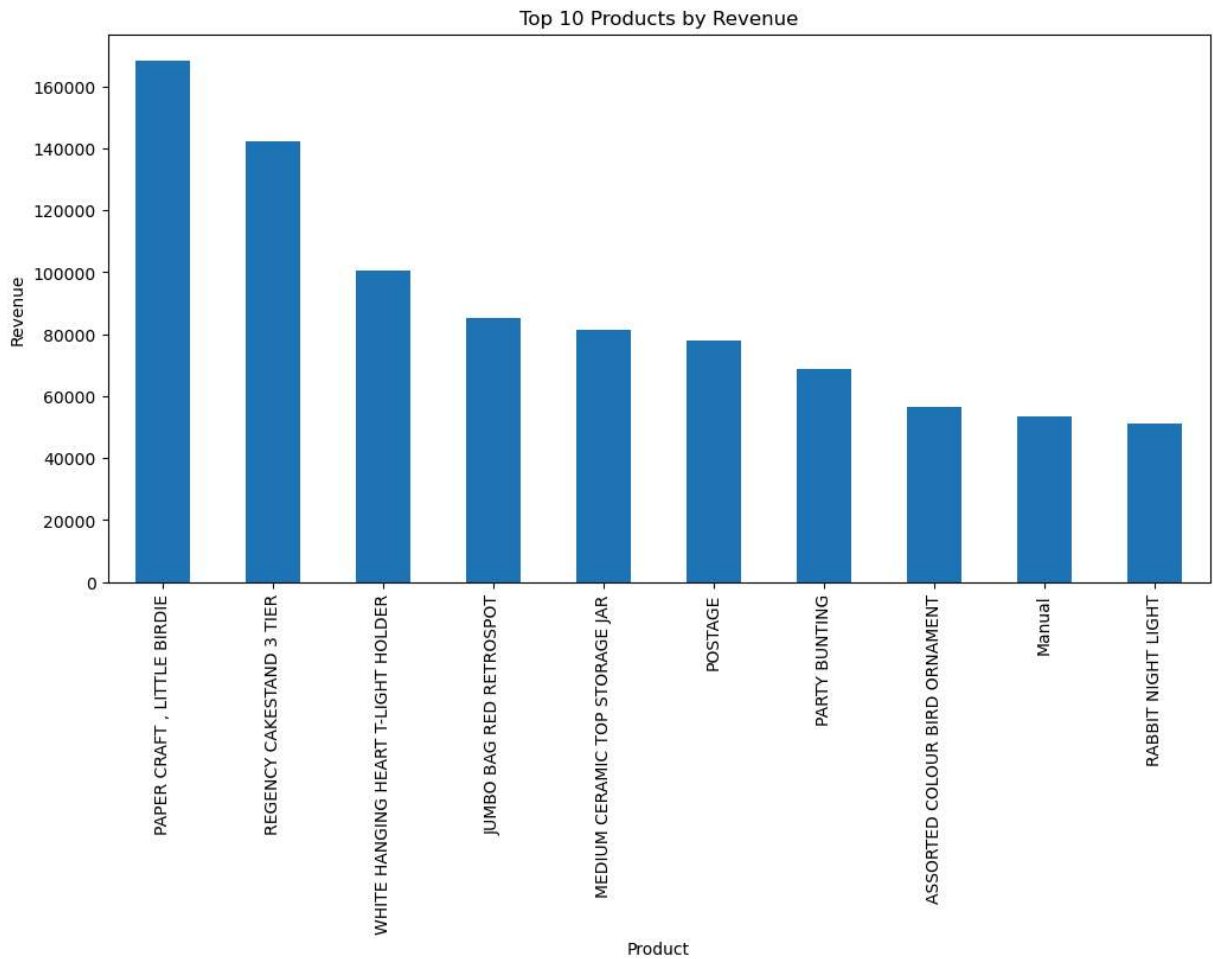
```
In [55]: plt.figure(figsize=(12,6))

product_revenue.head(10).plot(kind='bar')

plt.title("Top 10 Products by Revenue")
plt.xlabel("Product")
plt.ylabel("Revenue")
```

```
plt.xticks(rotation=90)

plt.show()
```



```
In [56]: df['Month'] = df['InvoiceDate'].dt.to_period('M')
```

```
In [57]: monthly_revenue = (
    df.groupby('Month')['TotalAmount']
    .sum()
)
```

```
In [58]: plt.figure(figsize=(12,6))

monthly_revenue.plot(marker='o')

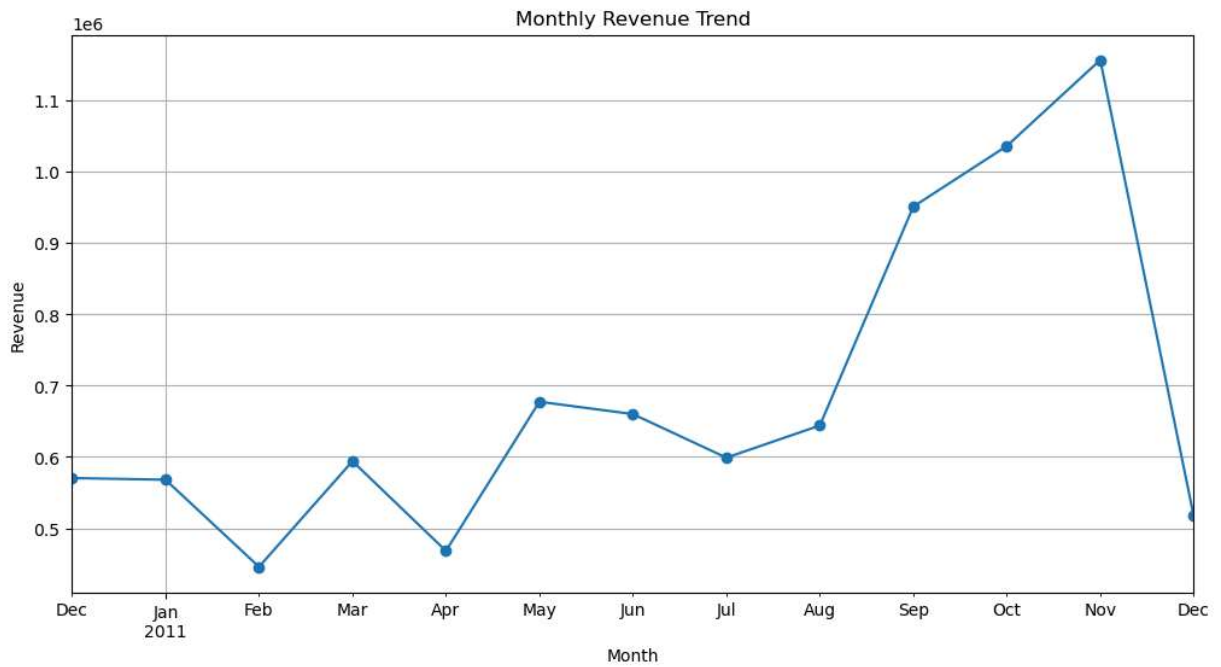
plt.title("Monthly Revenue Trend")

plt.xlabel("Month")

plt.ylabel("Revenue")

plt.grid(True)

plt.show()
```



```
In [66]: # top customers
customer_revenue = (
    df.groupby('CustomerID')['TotalAmount']
      .sum()
      .sort_values(ascending=False)
)
```

```
In [67]: customer_revenue.head(10)
```

```
Out[67]: CustomerID
14646.0      280206.02
18102.0      259657.30
17450.0      194390.79
16446.0      168472.50
14911.0      143711.17
12415.0      124914.53
14156.0      117210.08
17511.0       91062.38
16029.0       80850.84
12346.0       77183.60
Name: TotalAmount, dtype: float64
```

```
In [65]: plt.figure(figsize=(12,6))

customer_revenue.head(10).plot(kind='bar')

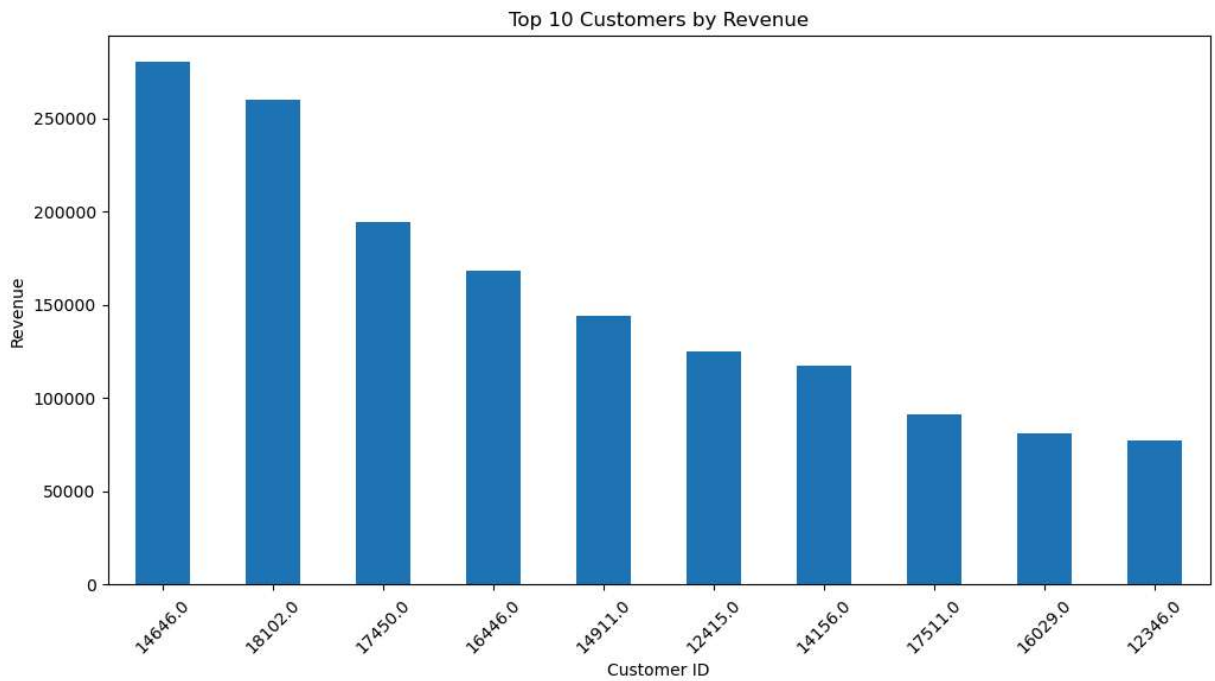
plt.title("Top 10 Customers by Revenue")

plt.xlabel("Customer ID")

plt.ylabel("Revenue")

plt.xticks(rotation=45)
```

```
plt.show()
```



```
In [68]: # purchase frequency
purchase_frequency = (
    df.groupby('CustomerID')['InvoiceNo']
      .nunique()
      .sort_values(ascending=False)
)
```

```
In [69]: purchase_frequency.head(10)
```

```
Out[69]: CustomerID
12748.0    209
14911.0    201
17841.0    124
13089.0     97
14606.0     93
15311.0     91
12971.0     86
14646.0     73
16029.0     63
13408.0     62
Name: InvoiceNo, dtype: int64
```

```
In [70]: plt.figure(figsize=(12,6))

purchase_frequency.head(10).plot(kind='bar')

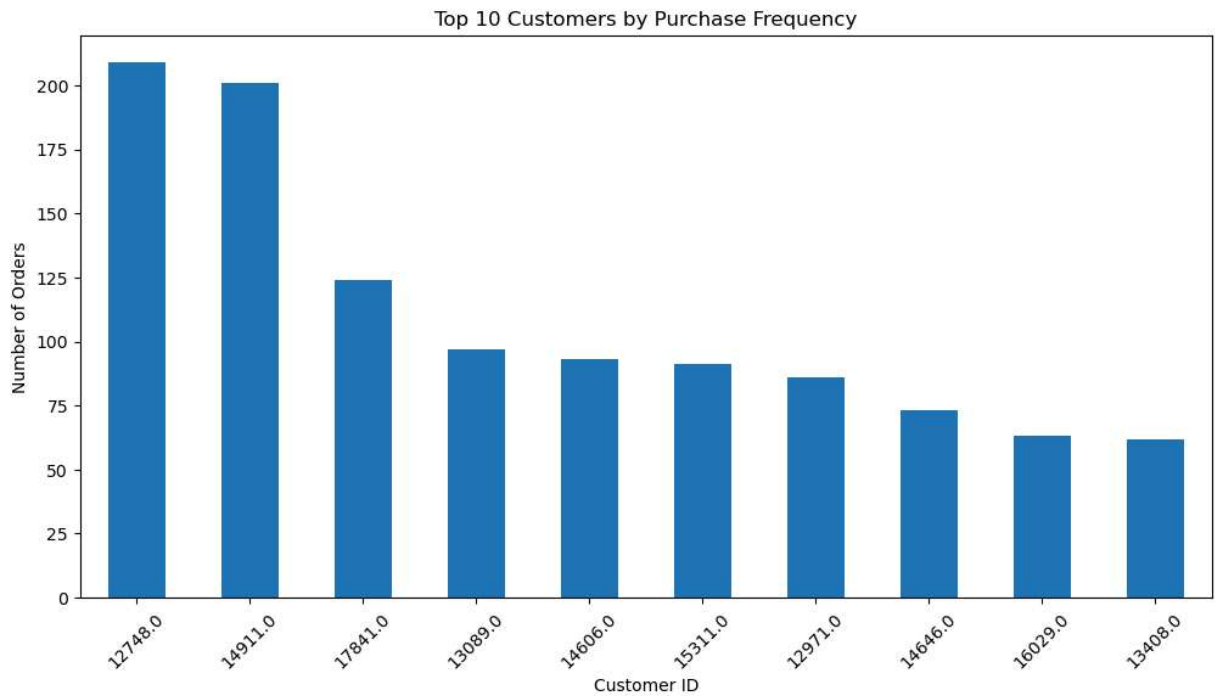
plt.title("Top 10 Customers by Purchase Frequency")

plt.xlabel("Customer ID")

plt.ylabel("Number of Orders")
```

```
plt.xticks(rotation=45)
```

```
plt.show()
```



```
In [71]: snapshot_date = df['InvoiceDate'].max() + pd.Timedelta(days=1)
```

```
In [75]: recency = (  
    df.groupby('CustomerID')['InvoiceDate']  
        .max()  
    )
```

```
In [76]: recency = (snapshot_date - recency).dt.days
```

```
In [79]: recency.head()
```

```
Out[79]: CustomerID  
12346.0    326  
12347.0     2  
12348.0    75  
12349.0    19  
12350.0   310  
Name: InvoiceDate, dtype: int64
```

```
In [80]: rfm = pd.DataFrame({  
    'Recency': recency,  
    'Frequency': purchase_frequency,  
    'Monetary': customer_revenue  
})
```

```
In [81]: rfm.head()
```

Out[81]:

	Recency	Frequency	Monetary
--	---------	-----------	----------

CustomerID			
12346.0	326	1	77183.60
12347.0	2	7	4310.00
12348.0	75	4	1797.24
12349.0	19	1	1757.55
12350.0	310	1	334.40

```
In [82]: rfm['R_Score'] = pd.qcut(
    rfm['Recency'],
    q=5,
    labels=[5,4,3,2,1]
)
```

```
In [84]: rfm['F_Score'] = pd.qcut(
    rfm['Frequency'],
    q=5,
    labels=False,
    duplicates='drop'
) + 1
```

```
In [85]: rfm['M_Score'] = pd.qcut(
    rfm['Monetary'],
    q=5,
    labels=[1,2,3,4,5]
)
```

```
In [86]: rfm.head()
```

Out[86]:

	Recency	Frequency	Monetary	R_Score	F_Score	M_Score
--	---------	-----------	----------	---------	---------	---------

CustomerID						
12346.0	326	1	77183.60	1	1	5
12347.0	2	7	4310.00	5	4	5
12348.0	75	4	1797.24	2	3	4
12349.0	19	1	1757.55	4	1	4
12350.0	310	1	334.40	1	1	2

```
In [87]: print(rfm.dtypes)
```

```
Recency      int64
Frequency    int64
Monetary     float64
R_Score      category
F_Score      int64
M_Score      category
dtype: object
```

```
In [88]: rfm['R_Score'] = rfm['R_Score'].astype(int)
rfm['M_Score'] = rfm['M_Score'].astype(int)
```

```
In [89]: print(rfm.dtypes)
```

```
Recency      int64
Frequency    int64
Monetary     float64
R_Score      int64
F_Score      int64
M_Score      int64
dtype: object
```

```
In [90]: rfm['RFM_Score'] = (
    rfm['R_Score'].astype(str) +
    rfm['F_Score'].astype(str) +
    rfm['M_Score'].astype(str)
)
```

```
In [91]: rfm.head()
```

```
Out[91]:
```

	Recency	Frequency	Monetary	R_Score	F_Score	M_Score	RFM_Score
--	---------	-----------	----------	---------	---------	---------	-----------

CustomerID							
------------	--	--	--	--	--	--	--

12346.0	326	1	77183.60	1	1	5	115
12347.0	2	7	4310.00	5	4	5	545
12348.0	75	4	1797.24	2	3	4	234
12349.0	19	1	1757.55	4	1	4	414
12350.0	310	1	334.40	1	1	2	112

```
In [92]: def customer_segment(row):

    if row['R_Score'] >= 4 and row['F_Score'] >= 4:
        return 'Champions'

    elif row['R_Score'] >= 3 and row['F_Score'] >= 3:
        return 'Loyal Customers'

    elif row['R_Score'] >= 4 and row['F_Score'] <= 2:
        return 'Potential Loyalists'

    elif row['R_Score'] <= 2 and row['F_Score'] >= 3:
        return 'At Risk'
```

```

else:
    return 'Lost Customers'

```

```
In [93]: rfm['Segment'] = rfm.apply(customer_segment, axis=1)
```

```
In [94]: rfm.head()
```

```
Out[94]:
```

	Recency	Frequency	Monetary	R_Score	F_Score	M_Score	RFM_Score	Seg
CustomerID								
12346.0	326	1	77183.60	1	1	5	115	Custo
12347.0	2	7	4310.00	5	4	5	545	Cham
12348.0	75	4	1797.24	2	3	4	234	A
12349.0	19	1	1757.55	4	1	4	414	Pot Lo
12350.0	310	1	334.40	1	1	2	112	Custo



```
In [98]: segment_count = rfm['Segment'].value_counts()

print(segment_count)
```

```

Segment
Lost Customers      2099
Potential Loyalists    737
Loyal Customers      721
Champions           578
At Risk             203
Name: count, dtype: int64

```

```
In [99]: import matplotlib.pyplot as plt

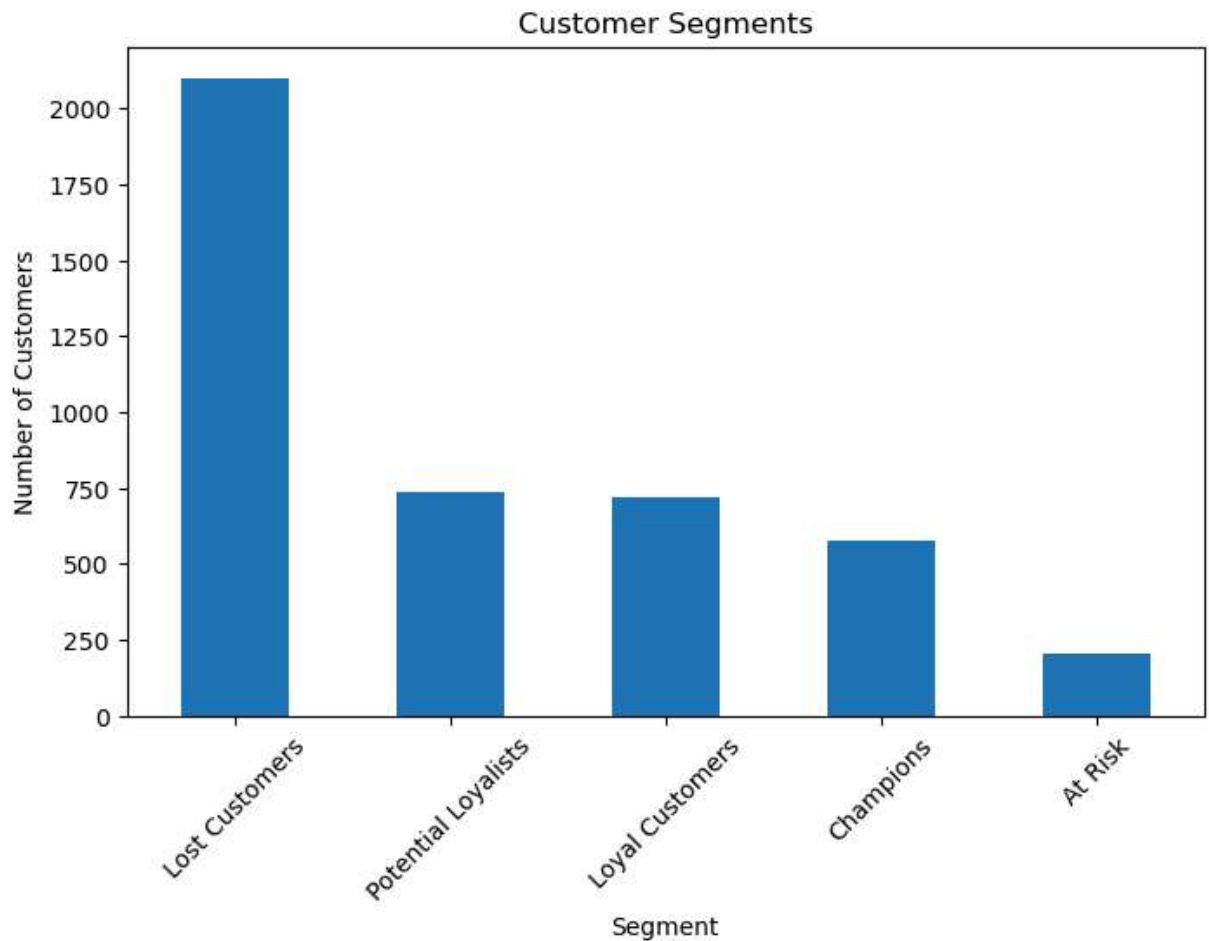
plt.figure(figsize=(8,5))

segment_count.plot(kind='bar')

plt.title("Customer Segments")
plt.xlabel("Segment")
plt.ylabel("Number of Customers")

plt.xticks(rotation=45)

plt.show()
```



```
In [103... segment_summary = rfm.groupby('Segment').agg(
    Avg_Recency=('Recency', 'mean'),
    Avg_Frequency=('Frequency', 'mean'),
    Avg_Monetary=('Monetary', 'mean'),
    Customer_Count=('Recency', 'count')
)

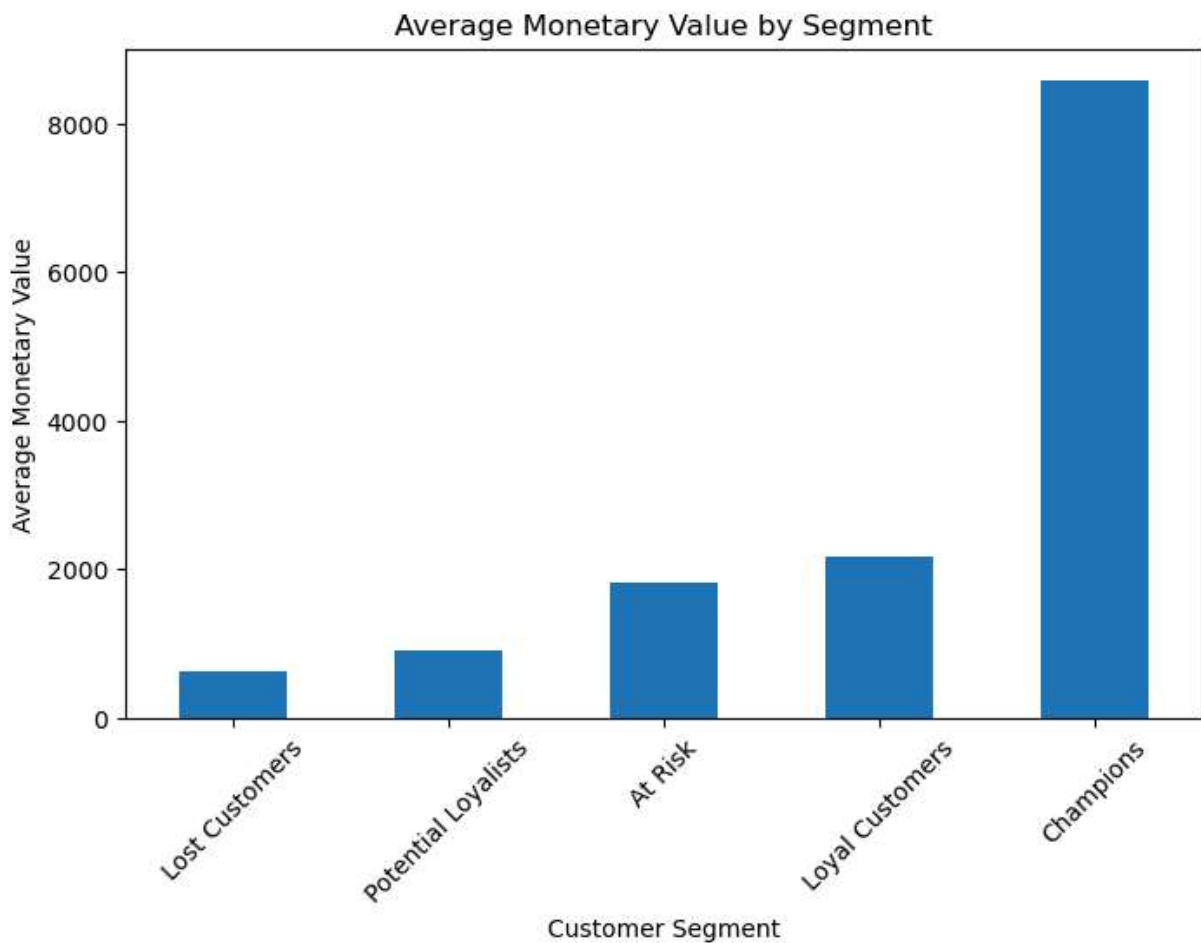
segment_summary
```

```
Out[103...      Avg_Recency  Avg_Frequency  Avg_Monetary  Customer_Count
Segment
At Risk      131.339901      5.561576    1826.597438          203
Champions     11.107266     15.335640    8573.205000          578
Lost Customers 159.692234      1.545974     631.435352         2099
Loyal Customers 28.011096      5.342580    2164.387200          721
Potential Loyalists 17.572592      1.956581     916.134532          737
```

```
In [104... rfm.columns
```

```
Out[104...] Index(['Recency', 'Frequency', 'Monetary', 'R_Score', 'F_Score', 'M_Score',  
      'RFM_Score', 'Segment'],  
      dtype='object')
```

```
In [105...] plt.figure(figsize=(8,5))  
  
segment_summary['Avg_Monetary'].sort_values().plot(kind='bar')  
  
plt.title("Average Monetary Value by Segment")  
plt.xlabel("Customer Segment")  
plt.ylabel("Average Monetary Value")  
  
plt.xticks(rotation=45)  
  
plt.show()
```

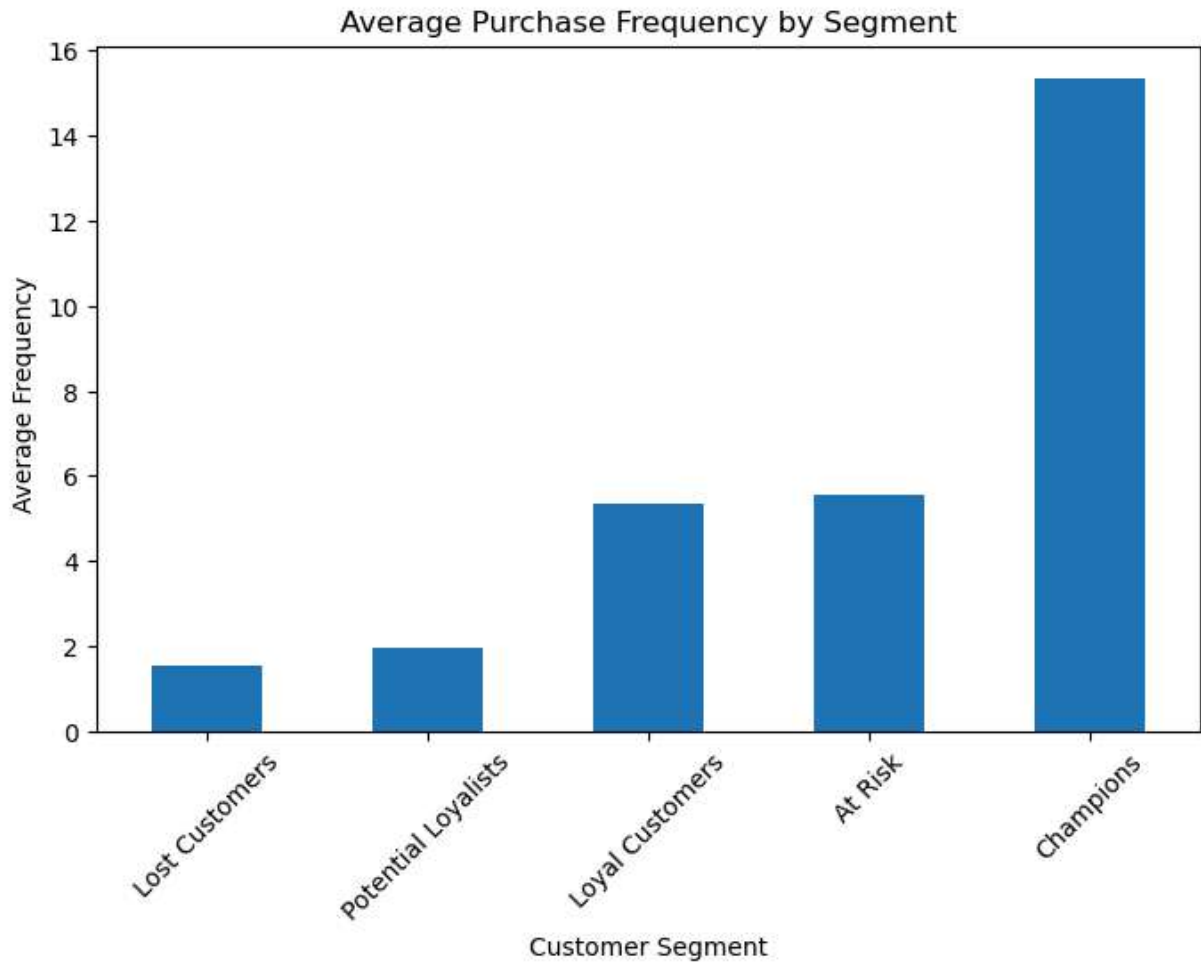


```
In [106...] segment_summary.columns
```

```
Out[106...] Index(['Avg_Recency', 'Avg_Frequency', 'Avg_Monetary', 'Customer_Count'], dtype='object')
```

```
In [110...] plt.figure(figsize=(8,5))  
  
segment_summary['Avg_Frequency'].sort_values().plot(kind='bar')  
  
plt.title("Average Purchase Frequency by Segment")  
plt.xlabel("Customer Segment")
```

```
plt.ylabel("Average Frequency ")
plt.xticks(rotation=45)
plt.show()
```



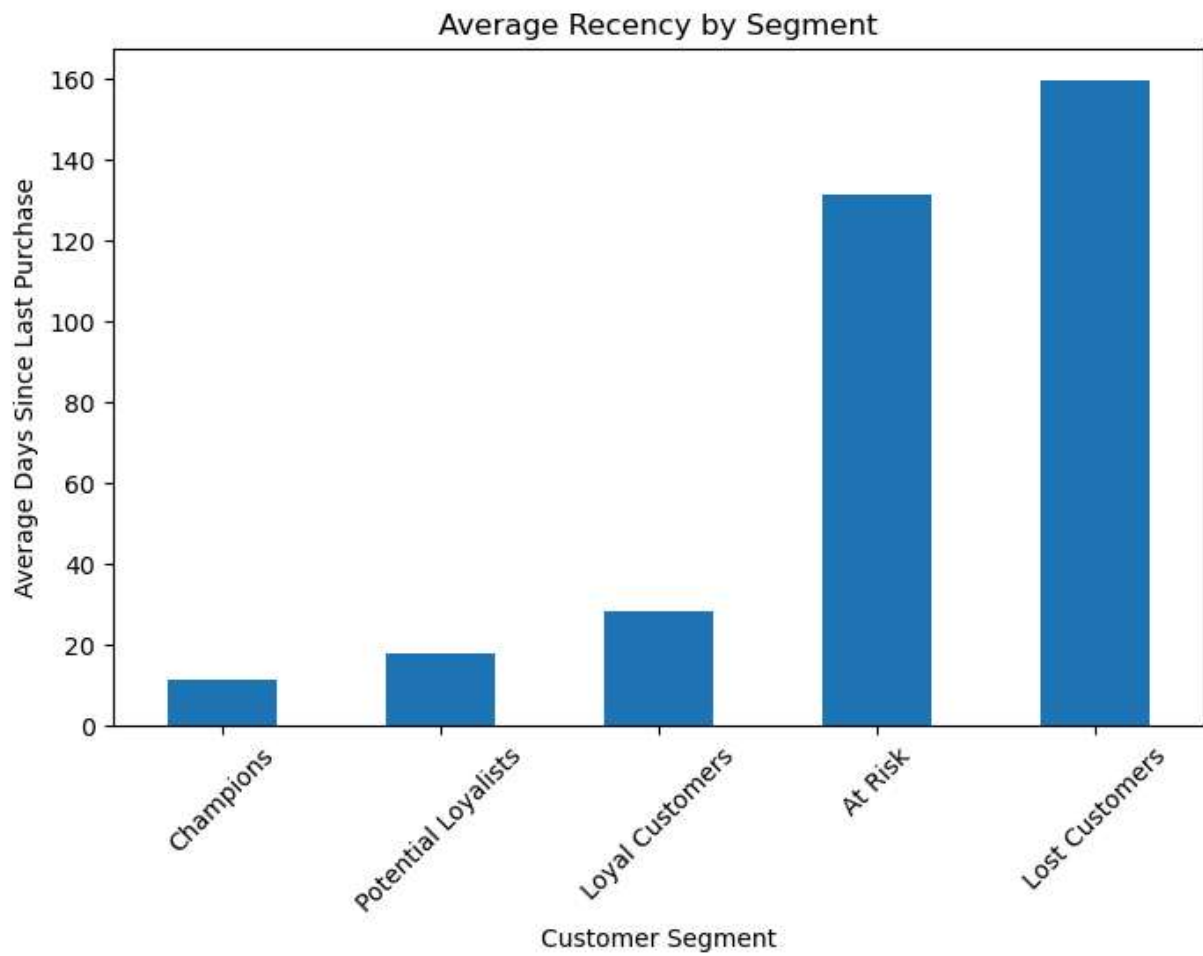
```
In [115... plt.figure(figsize=(8,5))

segment_summary['Avg_Recency'].sort_values().plot(kind='bar')

plt.title("Average Recency by Segment")
plt.xlabel("Customer Segment")
plt.ylabel("Average Days Since Last Purchase")

plt.xticks(rotation=45)

plt.show()
```



```
In [120...] segment_revenue = rfm.groupby('Segment')['Monetary'].sum()
print(segment_revenue.sort_values())
```

```
Segment
At Risk      370799.280
Potential Loyalists  675191.150
Lost Customers  1325382.803
Loyal Customers  1560523.171
Champions    4955312.490
Name: Monetary, dtype: float64
```

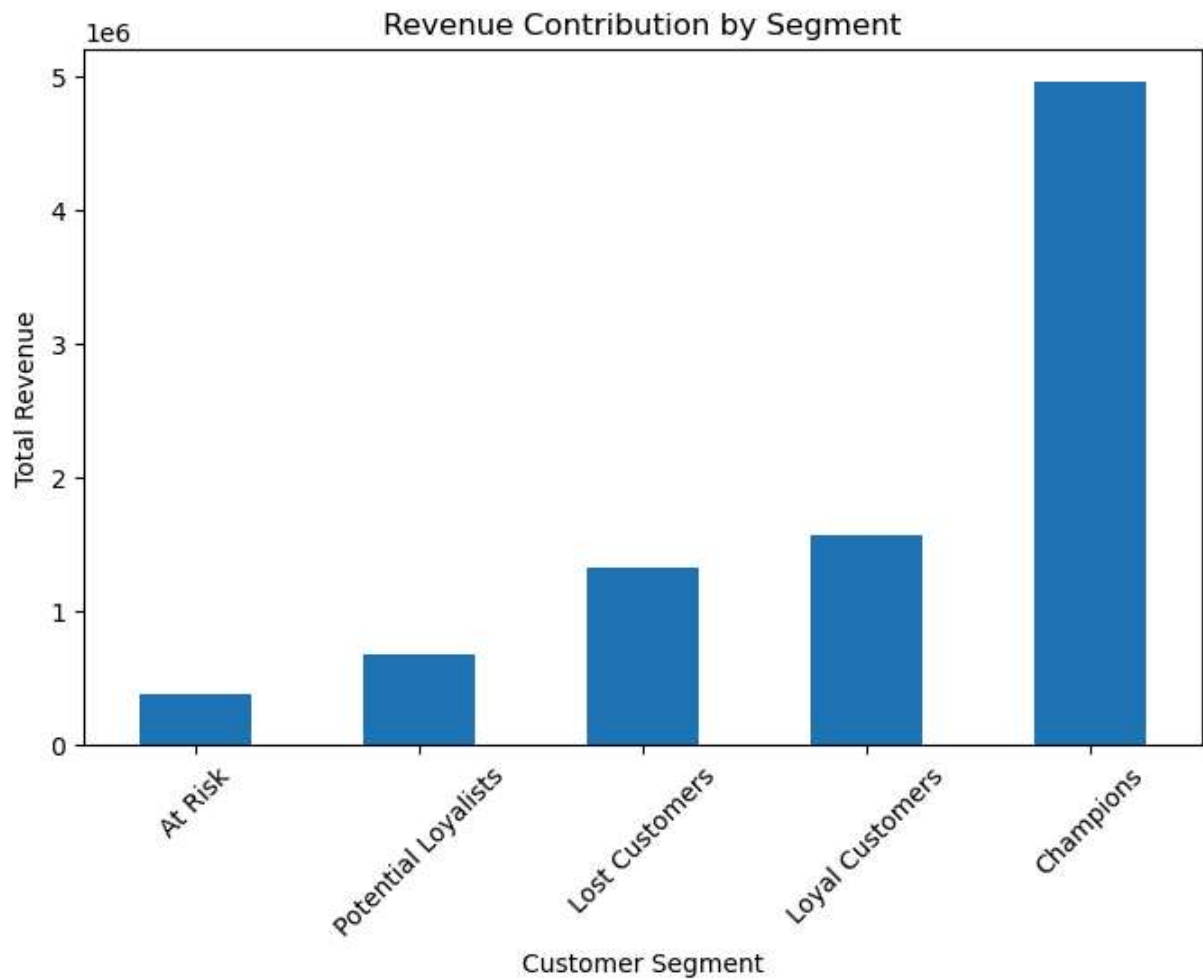
```
In [121...] plt.figure(figsize=(8,5))

segment_revenue.sort_values().plot(kind='bar')

plt.title("Revenue Contribution by Segment")
plt.xlabel("Customer Segment")
plt.ylabel("Total Revenue")

plt.xticks(rotation=45)

plt.show()
```



```
In [122...] rfm['Average_Order_Value'] = rfm['Monetary'] / rfm['Frequency']
```

```
In [123...] rfm['Average_Order_Value']
```

```
Out[123...] CustomerID
12346.0      77183.600000
12347.0       615.714286
12348.0       449.310000
12349.0      1757.550000
12350.0       334.400000
...
18280.0       180.600000
18281.0        80.820000
18282.0        89.025000
18283.0       127.845625
18287.0       612.426667
Name: Average_Order_Value, Length: 4338, dtype: float64
```

```
In [ ]:
```